

Vision2MeshAI

Complete Project Documentation — Version 2

AI-Based 2D / Multi-View Visual Input to 3D Reconstruction

Problem → Idea → Solution → Architecture → Dataset → Training → Results → Product Vision → Presentation

Current verified prototype: front/side/top views → CNN feature extraction → multi-view fusion → 64×64×64 voxel reconstruction.

Status note: blueprint/sketch/anime/general drawing support, production mesh cleanup, and a print-ready STL workflow are broader goals and are not claimed here as fully verified end-to-end features.

1. Project Overview

Vision2MeshAI is an AI-based 3D reconstruction project. Its central goal is to learn the relationship between 2D visual information and the hidden 3D structure of an object.

The current verified system uses three views — front, side and top — generated from known 3D meshes. Each view is processed by a CNN, the features are fused, and a 3D decoder predicts an occupancy voxel volume.

The broader product vision is to extend this foundation toward practical drawings, blueprints, toy/object representations and stylized images, followed by mesh reconstruction, visualization and 3D-printing-oriented processing.

One-line idea: “Use AI to reconstruct a 3D representation of an object from its 2D visual information.”

2. Problem and Pain Point

A person may have an image, drawing or blueprint of an object but not a ready-made 3D model. Creating the model manually can require time, 3D modelling skills and specialized software.

- 2D images lose depth information.
- Hidden surfaces must be inferred.
- Thin structures, holes and gaps are difficult to recover.
- Different objects can have similar silhouettes from one view.
- Blueprints and sketches are not directly 3D meshes.
- Stylized/anime drawings can differ strongly from realistic object images.

Presentation problem statement: “How can AI reconstruct the 3D structure of an object from available 2D visual information and reduce the need for manual modelling?”

3. Project Idea

The project learns from paired examples: multiple 2D views of an object and its known 3D target. During training, the network learns which visual patterns correspond to shape and spatial structure.

The project starts with controlled multi-view data because this makes the core reconstruction problem measurable. More difficult input styles can then be added as later stages.

4. Complete Proposed Solution

1. Collect or generate suitable 2D object views.
2. Resize and normalize the images.
3. Extract features independently using CNN encoders.
4. Fuse front, side and top features.
5. Project the fused representation into 3D.
6. Decode it into a 64×64×64 occupancy volume.
7. Extract a surface mesh in the future mesh stage.
8. Clean, repair and validate the mesh.
9. Visualize and export the model.
10. Prepare the exported model for downstream 3D-printing workflows.

Pipeline: 2D Views → Pre-processing → CNN Encoder → Multi-View Fusion → 3D Voxel Decoder → 64³ Voxel Model → Mesh Reconstruction → Cleanup → Visualization / Export → Future 3D-Printing Workflow

5. Current Implementation vs Broader Vision

Component	Current verified	Broader vision
Dataset	ModelNet10, 4,899 meshes	More diverse real-world/drawing data
Input	Front / side / top	Blueprints, sketches, toy/object and stylized inputs
Encoder	CNN	Stronger/pretrained visual encoders
Fusion	Three-view fusion	Flexible view-aware fusion
3D output	32 ³ baseline + 64 ³ model	Higher-resolution/alternative representations
Mesh	Not fully verified end-to-end	Voxel-to-mesh extraction and cleanup
Printing	Future workflow	STL/OBJ, watertightness, printability
Application	Frontend/backend areas	Complete upload → inference → preview → export

6. AI Architecture

6.1 Image Encoder

The current encoder processes each view independently using convolutional blocks, batch normalization, ReLU and max pooling. With 224×224 input, the verified encoder produces 128 channels at 14×14 spatial resolution.

6.2 Multi-View Fusion

The front, side and top feature maps are concatenated and passed through convolutional layers that learn how the three directions complement one another.

6.3 3D Decoder

The fused representation is projected into a compact 3D latent representation and expanded through 3D transposed convolutions until a 64×64×64 occupancy volume is produced.

6.4 Why voxels?

A voxel is a small cube in 3D space. The target grid records occupied and unoccupied regions. This gives a fixed-size supervised 3D representation. The trade-off is computational cost as resolution increases.

7. Dataset

The local ModelNet10 source contains exactly **4,899 real meshes**. For each mesh, the project generates front, side and top PNG views and matching voxel targets.

- 4,899 front images
- 4,899 side images
- 4,899 top images
- 4,899 32³ voxel targets
- 4,899 64³ voxel targets

The 64³ target generation finished at **4,899 / 4,899** completed samples.

8. 32³ Baseline Results

Metric	Result
Epochs	30
Best validation epoch	19
Mean Dice	0.7455
Median Dice	0.7763

Metric	Result
Std Dice	0.1548
Mean IoU	0.6171
Median IoU	0.6344
Std IoU	0.1866
Parameters	3,419,241

8.1 Category results

Category	Dice	IoU
dresser	0.8842	0.8184
sofa	0.8381	0.7306
bed	0.8052	0.6972
toilet	0.8005	0.6853
monitor	0.7741	0.6443
night_stand	0.7007	0.5665
table	0.6756	0.5269
chair	0.6586	0.5025
desk	0.6481	0.4992
bathtub	0.6125	0.4536

Visual tests showed that global shapes were learned, but thin supports, openings and small details were often lost. This motivated the 64³ experiment.

9. 64³ Upgrade and Final Training

The 64³ representation provides more spatial positions than 32³ and was introduced to improve the representation of finer geometry.

Setting	Result
Samples	4,899
Train / validation / test	3,429 / 734 / 736
Input	224×224
Batch	1
CPU threads	8
Parameters	1,145,802
Training	15 / 15 epochs completed
Epoch 15 train loss	0.482259
Epoch 15 validation loss	0.628514
Best validation loss	0.627769
Epoch 15 time	1h 16m 16s
Validation time	1m 20s

Setting	Result
Total training time	10h 55m 5s
Best checkpoint	C:\Users\Sowmya\Desktop\Vision2MeshAI\checkpoints_64\full_64_best_15e

Important: the best validation loss was 0.627769, slightly better than the epoch-15 validation loss of 0.628514. The saved best checkpoint represents the lowest validation loss reached during the run.

10. Performance Engineering

Experiment	Observed result
4 CPU threads	1.776 sec/batch
6 CPU threads	1.623 sec/batch
8 CPU threads	1.568 sec/batch
10 CPU threads	1.718 sec/batch
12 CPU threads	1.785 sec/batch
512×512 input	1.494 sec/batch
224×224 input	0.348 sec/batch

The project selected 8 CPU threads and 224×224 input based on measured benchmarks on the development machine.

11. Evaluation Concepts

Dice: measures overlap between predicted and target occupied voxels.

IoU: measures intersection divided by union of predicted and target occupied regions.

Training/validation loss: measures how closely predictions match targets during training and validation.

The 32³ experiment has complete test metrics. A separate 64³ test-set Dice/IoU evaluation is still required before claiming final 64³ test performance.

12. Software Structure

Path	Purpose
C:\Users\Sowmya\Desktop\Vision2MeshAI\dataset	Raw/generated data
C:\Users\Sowmya\Desktop\Vision2MeshAI\models	Model components
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\encoder	Older image_encoder.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\fusion	Older fusion_fusion.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\reco_32	32 ³ detection_voxel_decoder.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\reco_64	64 ³ detection_voxel_decoder_64.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\multiscale_model	64 ³ model.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\multiscale_model_64	64 ³ model_64.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\training	Training scripts
C:\Users\Sowmya\Desktop\Vision2MeshAI\scripts	Dataset generation
C:\Users\Sowmya\Desktop\Vision2MeshAI\checkpoints_32	32 ³ checkpoint
C:\Users\Sowmya\Desktop\Vision2MeshAI\checkpoints_64	64 ³ checkpoint

Path	Purpose
C:\Users\Sowmya\Desktop\Vision2MeshAI\frontend	Frontend area
C:\Users\Sowmya\Desktop\Vision2MeshAI\backend	Backend/API area

13. Technology Stack

Technology	Role
Python	Project/ML development
PyTorch	Deep learning
NumPy	Numerical and voxel arrays
Pillow	Image processing
Trimesh	Mesh processing and voxelization
FastAPI	Backend/API direction
Vite + React	Frontend direction
ModelNet10	Controlled training source

14. Broader Input Vision

14.1 Blueprint

Blueprints can contain orthographic views, dimensions and line geometry. A future system can combine these constraints with learned visual features.

14.2 Toy/object drawing

Drawings provide shape cues but may not contain exact depth or dimensions. Dedicated training examples would be required.

14.3 Anime/stylized object

Stylized imagery can distort perspective and hide geometry. Domain-specific training and evaluation would be needed.

14.4 Single image

The current verified model uses three views. A single-image mode would be a separate extension requiring missing-view inference or direct single-image-to-3D learning.

These are future product directions, not current performance claims.

15. Mesh Reconstruction and 3D-Printing Vision

1. Extract a surface from voxels.
2. Remove disconnected noise.
3. Repair holes and improve surface continuity.
4. Validate normals and topology.
5. Check watertight/manifold properties.
6. Optionally check wall thickness and printability.
7. Export STL/OBJ.
8. Provide interactive 3D preview.

The current verified neural output is a voxel volume. A reliable production print workflow still requires these downstream geometry stages.

16. Final User Workflow

Stage	User experience
1. Upload	Select one or more object images/drawings.
2. Preprocess	System checks and normalizes the input.
3. AI reconstruction	Model predicts 3D occupancy.
4. 3D preview	User rotates and inspects the reconstruction.
5. Mesh repair	Future geometry stage cleans the model.
6. Export	STL/OBJ or supported format.
7. Print preparation	Future printability and fabrication checks.

17. What the Project Has Demonstrated

- 4,899 ModelNet10 meshes available.
- Complete paired front/side/top image generation.
- Complete 32^3 target generation.
- Complete 64^3 target generation.
- 32^3 model trained and quantitatively tested.
- 32^3 visual reconstruction tests.
- 64^3 model and training pipeline verified.
- Full 15-epoch 64^3 training completed.
- Best 64^3 checkpoint saved.
- CPU and resolution benchmarks completed.

18. Current Limitations

- Controlled ModelNet10 data rather than broad real-world data.
- Current verified model expects three views.
- 64^3 training is expensive on CPU.
- Fine geometry remains challenging.
- Final 64^3 test Dice/IOU has not yet been established here.
- General blueprint/sketch/anime support is not verified.
- Production mesh cleanup and printability validation are not yet verified end-to-end.

19. Future Scope

- Evaluate the best 64^3 checkpoint on the 736-sample test set.
- Compare 32^3 and 64^3 visual reconstructions.
- Expand to real images and drawing styles.
- Develop single-image reconstruction.
- Improve fine-detail recovery.

- Experiment with stronger encoders or alternative 3D representations.
- Implement voxel-to-mesh extraction.
- Add mesh repair and watertightness checks.
- Add STL/OBJ export and 3D preview.
- Add printability and wall-thickness analysis.
- Integrate the complete frontend/backend workflow.

20. Full Presentation Explanation

Problem

“Most object information is available as a 2D image or drawing, but many applications need a 3D model. Manual modelling takes time and skill, and a 2D image does not directly provide complete depth.”

Idea

“Our idea is Vision2MeshAI — an AI system that learns the relationship between 2D visual information and 3D object structure.”

Solution

“Our current prototype uses front, side and top views. A CNN extracts features from each view, we fuse them, and a 3D decoder predicts the occupied 3D space as a voxel model.”

Dataset

“We use ModelNet10. Our local dataset contains 4,899 meshes. From each mesh we generate three views and corresponding voxel targets.”

32³ result

“The 32³ baseline achieved a mean Dice of 0.7455 and mean IoU of 0.6171. Visual tests showed good global shape learning but loss of some fine geometry.”

64³ result

“We generated all 4,899 64³ targets and completed 15 epochs of training. The best validation loss was 0.627769, and the best checkpoint was saved.”

Final vision

“Our broader goal is to support practical drawings and visual inputs, reconstruct a clean mesh, visualize it and prepare it for export and future 3D-printing workflows.”

21. 60-Second Presentation Script

“Our project is Vision2MeshAI, an AI-based 2D-to-3D reconstruction system. The problem is that people can have an image or drawing of an object but not a ready-made 3D model. Manual modelling takes time and skill, while a 2D image does not directly provide complete depth. Our current prototype takes front, side and top views. A CNN extracts features from each view, we fuse the three feature representations, and a 3D decoder predicts a voxel representation. We used 4,899 ModelNet10 meshes. Our 32³ baseline achieved a mean Dice of 0.7455 and mean IoU of 0.6171. We then generated all 4,899 64³ targets and completed 15 epochs of training, with a best validation loss of 0.627769. Our broader goal is to extend this toward practical drawings and other visual inputs, generate a clean 3D mesh and eventually prepare it for 3D-printing workflows.”

22. Viva Questions and Answers

Q: Why three views?

A: They provide complementary information and reduce the ambiguity of a single projection.

Q: Why AI?

A: The relationship between visual appearance and hidden 3D geometry is difficult to encode with fixed rules; the network can learn it from examples.

Q: Why ModelNet10?

A: It provides known 3D meshes, enabling controlled paired views and exact 3D supervision.

Q: Why voxels?

A: They provide a direct fixed-size 3D occupancy representation convenient for supervised learning.

Q: Why 64³?

A: It provides more spatial positions than 32³ and can represent finer geometry, at higher computational cost.

Q: Can it convert any drawing today?

A: Not yet. The verified model is trained on controlled three-view data.

Q: Can the current output be directly printed?

A: The current verified output is voxel data. Production mesh extraction, repair and printability stages are still needed.

Q: What did 32³ teach you?

A: It learned global shape but lost some thin structures, gaps and small details.

Q: What is the final 64³ result?

A: 15 epochs completed successfully; best validation loss was 0.627769. A separate test-set metric evaluation is still needed for final 64³ Dice/IoU claims.

Q: What is next?

A: Evaluate the best checkpoint on the test set, create visual comparisons, then continue the mesh and product pipeline.

23. Project Status

Item	Status
ModelNet10 source	Verified
4,899 three-view image sets	Verified
4,899 32 ³ targets	Verified
4,899 64 ³ targets	Verified
32 ³ training	Completed
32 ³ test metrics	Completed
32 ³ visual tests	Completed
64 ³ model verification	Completed
64 ³ 15-epoch training	Completed
64 ³ best checkpoint	Saved
64 ³ test Dice/IoU	Next evaluation
General blueprint input	Future
General sketch/anime input	Future
Single-image mode	Future
Production mesh cleanup	Future / validation required
Print-ready workflow	Future / validation required

24. Final Project Message

Vision2MeshAI is a staged AI reconstruction project. Its verified foundation is a supervised multi-view-to-3D voxel system trained using 4,899 ModelNet10 meshes.

The 32³ experiment established meaningful global reconstruction. The 64³ experiment increased target resolution and completed a full 15-epoch training run, with a best validation loss of 0.627769.

The long-term goal is to bridge the gap between 2D visual designs and usable 3D digital models. The next stages are broader input support, stronger fine-detail reconstruction, reliable mesh generation and a validated export/3D-printing workflow.

One-line project story: "Vision2MeshAI learns from 2D visual evidence to reconstruct the 3D structure of objects, with the long-term goal of turning visual designs into usable 3D models."

Appendix — Important Project Paths

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C:\Users\Sowmya\Desktop\Vision2MeshAI
C:\Users\Sowmya\Desktop\Vision2MeshAI\dataset\raw\ModelNet10
C:\Users\Sowmya\Desktop\Vision2MeshAI\dataset\mesh_supervised
C:\Users\Sowmya\Desktop\Vision2MeshAI\dataset\mesh_supervised_64
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\encoders\image_encoder.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\fusion\multiview_fusion.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\reconstruction\voxel_decoder.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\reconstruction\voxel_decoder_64.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\multiview_model.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\models\multiview_model_64.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\training\train_new_model.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\training\train_64_model.py
C:\Users\Sowmya\Desktop\Vision2MeshAI\checkpoints_new\best_model.pth
C:\Users\Sowmya\Desktop\Vision2MeshAI\checkpoints_64\full_64_best_15epoch.pth
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